LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety precautions to be observed during Gas/ Arc welding, Gas cutting and hot working
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety precautions to be observed during Gas/ Arc welding, Gas cutting and hot working
5. Specific Objectives	31.1.4.1 Trainee will be able to explain arc welding and brazing procedure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of PPE usage and inspection of welding equipment; Instructor supervision of safety protocol adherence
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary hazard associated with arc welding?
(b) List three essential Personal Protective Equipment items for gas cutting.
(c) Why is ventilation critical during hot working operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Personal Protective Equipment requirements • Fire prevention and extinguishing measures • Ventilation and fume extraction systems • Electrical safety during arc welding • Handling of oxygen and acetylene cylinders	L & Ds	PPT, IB	15 min
(b)	• Safe storage and transport of gases • Inspection of welding torches and hoses • Emergency shutdown procedures • First aid for burns and electric shock • Housekeeping and workspace organization	L & Ds	PPT, PDF	20 min
(c)	• Regulatory compliance and safety standards • Signage and warning labels usage • Hot work permit requirements • Coordination with other workshop activities • Post-operation equipment shutdown	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct procedure for checking gas cylinder valves before use.
(b) Describe the immediate actions required in case of a welding arc flash.
(c) Why must hot work permits be issued before starting gas cutting operations?

12. Home Assignment and Follow Up	Prepare a checklist of safety precautions for setting up an arc welding station, referencing the procedures to be covered in LP 10.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Equipment setup, tools and spares for Arc welding and procedure	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min